1) Intro

A game is a multi-person multi-objective problem. The models are usually simple.

A player knows the rules and is rational
A rational player does the following:

- Is **selfish**: acts for their own good
- **Aware of consequences**

Decision Problem

A decision problem consists of three main elements:

- **actions:** belong to a set A
- **outcomes:** is the result of the actions
- **preferences:** describes what is the preferred outcome

Preference

A preference is a binary relationship $\succcurlyeq$ on set A. If $a, b \in A$ and $a \succcurlyeq b$ then a is ranked above b.
We have two properties; a preference is said to be:

- **Complete:** if $\forall a, b \in A$ a preference is defined
- **Transitive:** if $\forall a, b, c \in A$ there exists an order: $a \succcurlyeq b \cap b \succcurlyeq c \implies a \succcurlyeq c$
  If $\succcurlyeq$ is complete and transitive then it is called **rational**

Utilities

Utilities (payoff functions) are an arbitrary quantification $u(q)$ of the goodness coming from some input q
If q is a countable good, $u(q)$ is generally an increasing function of q.
The exact formulation of $u(\cdot)$ does not matter, it just maps the order via $\geq$ on numbers.

A preference $\succcurlyeq$ can be put in a relationship with $u : A \to \mathbb{R}$
u **represents** $\succcurlyeq$ if

$$\forall a, b \in A \quad a \succcurlyeq b \iff u(a) \geq u(b)$$

Theorem 1 (Representation of $\succcurlyeq$ and $u(\cdot)$)**.**

On a finite set A, $\succcurlyeq$ can be represented by u $\iff$ $\succcurlyeq$ is rational

Quick Proof
$\implies$ *: is immediate (due to property of $\geq$)*
$\impliedby$ *: A suitable utility function can be $u(a) = |\{b \in A : a \succcurlyeq b\}|$*

Decision Trees

TODO

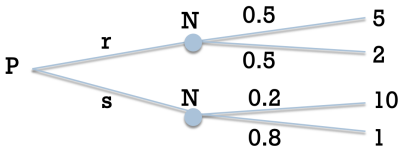
2) Lotteries

Randomness messes with rationality as it is harder to infer consequences

Example:

$$\text{Ravioli gives } u(r) = \begin{cases} 2 & w.p.\ 0.5 \\ 5 & w.p.\ 0.5 \end{cases}$$
$$\text{While Soup gives } u(s) = \begin{cases} 1 & w.p.\ 0.8 \\ 10 & w.p.\ 0.2 \end{cases}$$

Usually randomness is given from player actions (one player can also be nature)
Nature is **not** a rational player, its choices are iid as seen in the picture.



Tree

With $N \to \infty$ trials then payoff = expectation, where expectation is

$$\mathbb{E}[u(x)|p] = \sum_k p(x_k)u(x_k)$$

In this case we have that r is rationally preferable since $\mathbb{E}[u|r] = 3.5 \geq \mathbb{E}[u|s] = 2.8$.

Lottery

A lottery over outcomes $X = \{x_1, \dots, x_n\}$ is defined as a probability distribution p over X, that is

$$p = \{p(x_1), \dots, p(x_n)\} \text{ where } p(x_k) \in [0,1] \text{ and } \sum_k p(x_k) = 1$$

if actions are involved p is conditional $p(x_k|a)$

That is, a lottery specifies the probability of a said outcome. It can be conditional and it becomes the probability of an outcome given the previous action.

Expected Utility

We want to define $\succcurlyeq$ among lotteries, we replace A with the set $P(A)$ of lotteries over A

Axiom 2 (Continuity Axiom)**.**

For $p, q, r \in P(A)$ it must hold that the following sets are closed

$$\{a \in [0,1] : ap + (1-a)q \succcurlyeq r\}$$
$$\{a \in [0,1] : r \succcurlyeq ap + (1-a)q\}$$

That is, arbitrarily small variations in gamble does not change preferred lotteries

If you mix the best outcome (p) and the worst outcome (q) with a probability a, as long as a is high enough, you will prefer that gamble to the sure thing (r).

Axiom 3 (Independence Axiom)**.**

For $p, q, r \in P(A)$ it holds that $\forall a \in [0,1]$ if $p \succcurlyeq q$ then $ap + (1-a) \succcurlyeq aq + (1-a)r$
That is, when mixing gambles we prefer the preference order not counting other alternatives

Theorem 4 (Von Neumann & Morgenstern)**.**

If $\succcurlyeq$ satisfies the four axiums (complete, transitive, continuity and independence) it can be represented by:

$$u(\cdot) \text{ such that } \forall p, q \in P(A),\ p \succcurlyeq q \implies \mathbb{E}[u(x)|p] \geq \mathbb{E}[u(x)|q]$$

This function is called vN-M utility

vN-M does not state to compare expectations but to use affine transformations of u. That is a combination of degenerate lotteries $p = p_1(1,0,0,\dots) + p_2(0,1,0,\dots) + \dots$

Continuous Case

For example, dig a well to extract water, what is the preferred depth?
The depth is continuous: $0 \leq d \leq \text{Eart radius}$
The effort to dig is (imposed by problem): $d^2/2$
The extracted water is uniformly distributed: $W(d) \sim u[0, 20d]$

$$\mathbb{E}[u|d] = \mathbb{E}[W(d) - d^2] = 10d - \frac{d^2}{2}$$

The optimal value is therefore $d = 10 \to u = 50$.

Risk Attitude

Suppose three possible outcomes:

$$x_1 = 0 \quad x_2 = 1 \quad x_3 = 20$$

and two lotteries

$$p_A = (0, 1, 0) \qquad p_B = (0.95, 0, 0.05)$$

Both expected utilities are the same, how does one player chose which one is better?

Risk Neutral:
A risk neutral player will see these two choices as equivalent. This is a **linear** utility
Risk Averse:
A risk averse player prefers the degenerate lottery (A) as it guarantees the highest minimum win. This is a **concave** utility
Risk Loving:
A risk loving player prefers the lottery that allows for the max possible win. This is a **convex** utility

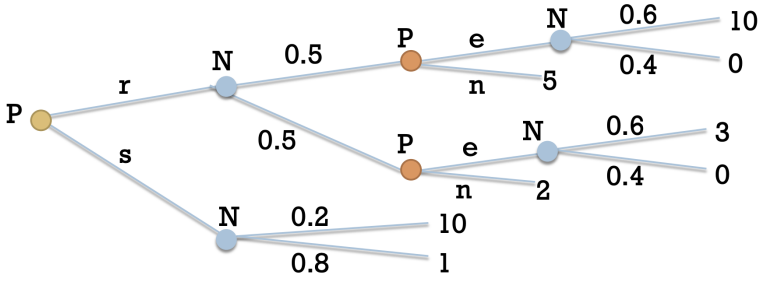
Therefore monotonic utilities do not change the preference, but change the risk attitude

Backwards Induction

Suppose that an entire tree is drawn with both player and nature choices. The solution (optimal decision) is found ia backwards induction

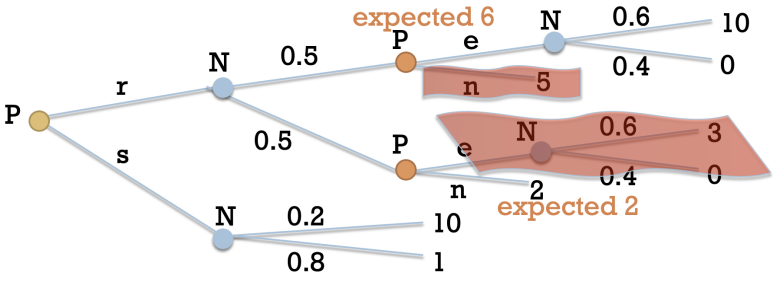
Classify all nodes with P action into groups:

- Group 1: all nodes after which no further action is possible in any case
- Group k: includes all nodes that are followed only by at least one Group $k-1$ node and no higher order node
  Start from group 1: find rational choice that maximizes utility, then go to group 2, repeat until last group is complete.



Example

Here Orange is group 1, yellow group $k = 2$.
Now find optimal decision for group 1 and remove the unused branches



Example Continued

In the pruned tree we notice that the optimal decision is r: $\mathbb{E}[u|r] = 4 \geq \mathbb{E}[u|s] = 2.8$

Value of Information

Now suppose a problem is solved with optimal utility $\mathbb{E}[u] = k$.
And suppose that now the nature choice is already known, then the optimal utility can increase $\mathbb{E}[u|\text{knowledge}] = k' \geq k$ as the uncertainties are removed.
The value of this information is $k' - k$

3) Static Games of Complete Information

With multiple players things change. All players want to maximize their payoff!

We consider the following game:

- static: all players move together (simultaneous and independent)
- complete information: anybody's payoff function is known

The combination of the choices determines the outcome of the game $(a_1, \dots, a_n)$, the utility of i-th player is

$$u_i = u_i(a_1, a_2, \dots, a_n)$$

An information E is common knowledge if:

- Everybody knows E
- It is known that everybody knows it

Normal Form Representation

Each player i has a set of accessible pure strategies S_i available to them.
We call the choice of ith player $s_i \in S_i$.
Each player has an utility u_i

We call the strategy profile all the actions taken by all players, that is

$$s = (s_1, \dots, s_i, \dots, s_n)$$

We call the strategy profile that excludes i-th player choice as

$$s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$$

The normal form of the game is

$$G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$$

If the sets are discrete, this can be written in a n dimensional matrix

- **Player A has three strategies: {U, M, D}**
- **Player B has two strategies: {L, R}**

		player B	
		L	R
player A	U	8, 0	0, 5
	M	1, 0	4, 3
	D	0, 7	2, 0

2 player Example

Dominance

A joint strategy s is **Pareto Dominated** by another strategy s' if

$$\begin{aligned} u_i(s') &\geq u_i(s) \quad \text{for all players } i \\ u_i(s') &> u(s) \quad \text{for some players } i \end{aligned}$$

s' is said to be **pareto efficient**

we say that s_i is **strictly dominated** by s_i' if the payoff of s_i' is always greater than s_i

$$u_i(s_{-i}, s_i') > u_i(s_{-i}, s_i) \quad \forall s_{-i}$$

rational players do not play s and they can be removed from the table.
This process is called **Iterated Elimination of Strictly Dominated Strategies (IESDS)**

Consider this example:

	L	R
U	$6,0$	$0,5$
M	$1,0$	$4,3$
D	$0,7$	$2,0$

So $S_A = \{U, M, D\},\ S_B = \{L, R\}$
Notice that for player A, D is dominated by M since

$$u_A(M, L) > u_A(D, L) \quad u_A(M, R) > u_A(D, R)$$

Now

	L	R
U	$6,0$	$0,5$
M	$1,0$	$4,3$

Now for player B, R dominates L since

$$u_B(U, R) > u_A(U, L) \quad u_B(M, R) > u_B(M, L)$$

So

	R
U	$0,5$
M	$4,3$

Now finally for player A, M dominates U

	R
M	$4,3$

rational player play (M, R).

4) Rationalizing Solutions

Strategy $s_i \in S_i$ is the best response to $s_{-i} \in S_{-i}$ if $u(s_i, s_{-i}) \geq u(s_i', s_{-i})\ \forall s_i' \in S_i$

p

Nash Equilibrium
Loop of dominant decisions from multiple players: x dominates y, so I play x, my opponent knows I'll do that and if y dominates x he'll play y.

Everybody happy!

Motivation
If this is not in a nash equilibrium then there exists some player i such that s_i' is not the best response
So there is an incentive for player i to change form s_i'